

LLD Design Patterns Priority for Freshers (SDE/SWE)

Tier1: Singleton, Strategy, Factory Method/Simple Factory, Observer, Builder.

Tier2: Adapter, Decorator, Command, State, Template Method.

Tier3: Facade, Proxy, Chain of Responsibility, Composite, Iterator, Bridge.

Tier4: Abstract Factory, Prototype, Flyweight, Mediator, Visitor, Memento, Interpreter.

Learning Order:

1.Singleton 2.Factory 3.Builder 4.Strategy 5.Observer 6.Adapter 7.Decorator 8.State 9.Command 10.Template Method 11.Facade 12.Proxy 13.Chain of Responsibility 14.Composite 15.Iterator.

Most used in practice: Strategy, Factory, Builder, Observer, Singleton, Adapter, Decorator, State, Facade, Proxy, Command, Chain of Responsibility, Composite.

Typical interview mappings:

Parking Lot→Strategy, Factory, Singleton

Splitwise→Strategy, Factory, Observer

Snake & Ladder→Strategy, State

Elevator→State, Strategy, Observer

BookMyShow→Factory, Observer

Chess→Strategy, State

Tic-Tac-Toe→Strategy

Vending Machine→State

ATM→State, Chain of Responsibility

Notification Service→Observer

Logging Framework→Singleton, Factory, Chain of Responsibility

Payment Gateway→Strategy, Factory, Adapter

One-week plan:

Day1 SOLID, Day2 Singleton+Factory, Day3 Builder, Day4 Strategy, Day5 Observer, Day6 Adapter+Decorator, Day7 State + one LLD problem.

Recommendation: Master SOLID first, then Singleton, Factory, Builder, Strategy, Observer, Adapter, State, Decorator, Command, and practice classic LLD problems (Parking Lot, Splitwise, Vending Machine, Elevator, Tic-Tac-Toe, Logging Framework).